

1. 1번	2. 2번	3. 1번	4. 1번	5. 4번
6. 2번	7. 1번	8. 4번	9. 4번	10. 2번
11. 3번	12. 2번	13. 1번	14. 3번	15. 4번
16. 1번	17. 3번	18. 5번	19. 2번	20. 3번
21. 1번	22. 3번	23. 2번	24. 3번	25. 1번
26. 4번	27. 2번	28. 3번	29. 2번	30. 3번

1. $\ln x$ 는 $x=0$ 에서 $f(x), f'(x), f''(x), \dots$ 가 정의가 되지 않으므로 $x=0$ 에서 Taylor급수 전개가 불가능하다.

2. $(x-e)^2$ 항의 계수 : $\frac{f''(e)}{2!}$

$$\Rightarrow f'(x) = 2x \ln x + x$$

$$\Rightarrow f''(x) = 2(\ln x + 1) + 1 = 2\ln x + 3$$

$$\therefore \frac{f''(e)}{2!} = \frac{5}{2}$$

3. $f(x) = (5-x)^{-1}, f'(x) = (5-x)^{-2}, f''(x) = 2!(5-x)^{-3}, \dots$

$$f(x) = f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \dots$$

$$= 4^{-1} + 4^{-2}(x-1) + 4^{-3}(x-1)^2 + \dots$$

$$= \sum_{n=0}^{\infty} \frac{(x-1)^n}{4^{n+1}}$$

[다른 풀이]

$$\frac{1}{5-x} = \frac{1}{4-(x-1)} = \frac{1}{4\left(1-\frac{x-1}{4}\right)} = \frac{1}{4}\left(1 + \frac{x-1}{4} + \frac{(x-1)^2}{16} + \dots\right)$$

$$= 4^{-1} + 4^{-2}(x-1) + 4^{-3}(x-1)^2 + \dots$$

$$= \sum_{n=0}^{\infty} \frac{(x-1)^n}{4^{n+1}}$$

4. $f(x) = \sqrt[3]{x}$ 일 때, $x=8$ 에서 Taylor전개

$$\Rightarrow f(x) = f(8) + f'(8)(x-8) + \frac{f''(8)}{2!}(x-8)^2 + \dots$$

$$\Rightarrow \sqrt[3]{x} = 2 + \frac{1}{12}(x-8) + \dots$$

$$\Rightarrow \sqrt[3]{9} \approx 2 + \frac{1}{12}(9-8) = 2.083$$

5. $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$

$$\Rightarrow \sin 1 = 1 - \frac{1}{3!} + \frac{1}{5!} - \frac{1}{7!} + \dots$$

$$\Rightarrow \sin 1 \approx 1 - \frac{1}{3!} + \frac{1}{5!} = 0.84$$

6. $f(x) = e^{-2x} \sin x$

$$= \left[1 + (-2x) + \frac{(-2x)^2}{2!} + \frac{(-2x)^3}{3!} + \dots\right] \times \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots\right)$$

$$= \dots \left(\frac{1}{5!} - \frac{1}{3!} + \frac{2}{3}\right)x^5 + \dots$$

$$\text{즉, } x^5 \text{의 계수 } \frac{1}{5!} + \frac{1}{3} = \frac{41}{120}$$

$$\begin{aligned} 7. f(x) &= \frac{x^3}{x+2} = \frac{x^3}{2} \times \frac{1}{1+\frac{1}{2}x} \\ &= \frac{1}{2}x^3 \left(1 - \frac{1}{2}x + \frac{1}{4}x^2 - \frac{1}{8}x^3 + \dots\right) \\ &= \frac{1}{2}x^3 - \frac{1}{4}x^4 + \frac{1}{8}x^5 - \frac{1}{16}x^6 + \dots \end{aligned}$$

[다른 풀이]

$$f(x) = \frac{x^3}{x+2} = x^2 - 2x + 4 - \frac{8}{x+2} \quad (\because \text{다항식 나눗셈})$$

$$= x^2 - 2x + 4 - \frac{4}{1+\frac{x}{2}}$$

$$= x^2 - 2x + 4 - 4 \left(1 - \frac{1}{2}x + \dots - \frac{1}{64}x^6 + \dots\right)$$

$$= \dots - \frac{x^6}{16}$$

$$\begin{aligned} 8. \frac{1}{1+x+2x^2} &= \frac{1}{1-(-x-2x^2)} \\ &= 1 + (-x-2x^2) + (-x-2x^2)^2 + \dots \\ &= 1 - x + (-2+1)x^2 + (4-1)x^3 + \dots \\ \text{즉, } a_0 + a_3 &= 4 \end{aligned}$$

$$9. f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

$$= a_0 + a_1x + a_2x^2 + a_3x^3 + \dots = \sum_{n=0}^{\infty} a_nx^n$$

$$\text{즉, } a_0 = f(0) = 1$$

10. $f(x) = e^{\sin x}$

$$f'(x) = e^{\sin x} \cos x$$

$$f''(x) = e^{\sin x} \cos^2 x - e^{\sin x} \sin x$$

$$f'''(x) = e^{\sin x} \cos^3 x - e^{\sin x} 2\cos x \sin x - e^{\sin x} \cos x \sin x - e^{\sin x} \cos x$$

$$\Rightarrow f(0) = 1, f'(0) = 1, f''(0) = 1, f'''(0) = 0$$

$$\Rightarrow f(x) = e^{\sin x} = 1 + x + \frac{1}{2}x^2$$

[다른 풀이]

$$f(x) = e^{\sin x}$$

$$= 1 + \sin x + \frac{1}{2!}\sin^2 x + \frac{1}{3!}\sin^3 x + \dots$$

$$= 1 + \left(x - \frac{1}{3!}x^3 + \dots\right) + \frac{1}{2!}\left(x - \frac{1}{3!}x^3 + \dots\right)^2$$

$$+ \frac{1}{3!}\left(x - \frac{1}{3!}x^3 + \dots\right)^3 + \dots$$

$$\Rightarrow 1 + x - \frac{1}{6}x^3 + \frac{1}{2}x^2 + \frac{1}{6}x^3 = 1 + x + \frac{1}{2}x^2$$

11. $f(x) = (\tan^{-1}x)^2$

$$= \left(x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \dots\right) \left(x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \dots\right)$$

$$= \dots + \left(\frac{1}{5} + \frac{1}{9} + \frac{1}{5}\right)x^6 + \dots$$

$$\text{즉, } f^{(6)}(0) = \frac{23}{45}6! = 368$$

$$\begin{aligned}
 12. f(x) &= \ln\left(\frac{1+x^2}{1-x^2}\right) = \ln(1+x^2) - \ln(1-x^2) \\
 &= \left(x^2 - \frac{1}{2}x^4 + \frac{1}{3}x^6 - \frac{1}{4}x^8 + \frac{1}{5}x^{10} - \dots\right) \\
 &\quad - \left(-x^2 - \frac{1}{2}x^4 - \frac{1}{3}x^6 - \frac{1}{4}x^8 - \frac{1}{5}x^{10} - \dots\right) \\
 &= 2\left(x^2 + \frac{1}{3}x^6 + \frac{1}{5}x^{10} + \dots\right) \\
 \therefore \frac{f^{(10)}(0)}{10!} &= \frac{2}{5}
 \end{aligned}$$

$$\begin{aligned}
 13. f(x) &= x \ln(1+x^2) = x\left(x^2 - \frac{1}{2}x^4 + \frac{1}{3}x^6 - \dots\right) \\
 &= x^3 - \frac{1}{2}x^5 + \frac{1}{3}x^7 - \dots \\
 \therefore f^{(5)}(0) &= -\frac{5!}{2} = -60
 \end{aligned}$$

$$\begin{aligned}
 14. f(x) &= \sqrt[3]{1+x^2} = (1+x^2)^{\frac{1}{3}} \\
 &= 1 + \frac{1}{3}x^2 + \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)}{2!}x^4 + \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)}{3!}x^6 + \dots \\
 &= 1 + \frac{1}{3}x^2 - \frac{1}{9}x^4 + \frac{5}{81}x^6 + \dots \\
 \therefore f^{(6)}(0) &= \frac{5}{81} \cdot 6! = \frac{400}{9}
 \end{aligned}$$

$$\begin{aligned}
 15. f(x) &= \int_0^x e^{-t^2} dt = \int_0^x \left(1 - t^2 + \frac{t^4}{2!} - \frac{t^6}{3!} + \dots\right) dt \\
 &= \left[t - \frac{1}{3}t^3 + \frac{1}{10}t^5 - \dots\right]_0^x = x - \frac{1}{3}x^3 + \frac{1}{10}x^5 - \dots \\
 \therefore x=0 \text{에서 } 5\text{차 Taylor 다항식은 } &x - \frac{1}{3}x^3 + \frac{1}{10}x^5 \text{ 이다.}
 \end{aligned}$$

$$\begin{aligned}
 16. \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} &= \frac{1}{1!} - \frac{1}{3!} + \frac{1}{5!} - \frac{1}{7!} + \dots \\
 &= \frac{1}{1} - \frac{1}{6} + \frac{1}{120} - \frac{1}{5100} + \dots \\
 \text{여기서 } \left|\frac{1}{5100}\right| &\approx 0.0002 < 0.0005 \left(\frac{1}{2000}\right) \text{ 이므로} \\
 \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} &\approx \sum_{k=0}^2 \frac{(-1)^k}{(2k+1)!} = \frac{1}{1} - \frac{1}{6} + \frac{1}{120} \text{ 이다.} \\
 \therefore \text{구하는 최소의 정수 } N &\text{은 } 2 \text{ 이다.}
 \end{aligned}$$

$$\begin{aligned}
 17. \int_0^1 \cos(\sqrt{x}) dx &= \int_0^1 \left(1 - \frac{(\sqrt{x})^2}{2!} + \frac{(\sqrt{x})^4}{4!} - \frac{(\sqrt{x})^6}{6!} + \dots\right) dx \\
 &= \left[x - \frac{x^2}{4} + \frac{x^3}{72} - \dots\right]_0^1 = 1 - \frac{1}{4} + \frac{1}{72} - \dots \\
 &= 0.75 + 0.013 - \dots \\
 &\approx 0.76
 \end{aligned}$$

$$18. \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \text{ 이므로 } \sum_{n=0}^{\infty} \frac{x^n}{2^n} = \frac{1}{1-\frac{x}{2}} = \frac{2}{2-x} \text{ 이다.}$$

$$\begin{aligned}
 19. f(x) &= \sum_{n=0}^{\infty} (-1)^n \frac{1}{n!} x^n = \sum_{n=0}^{\infty} \frac{1}{n!} (-x)^n = e^{-x} \\
 \therefore f'(x) &= -e^{-x} \text{ 이므로 } f'(1) = -e^{-1} \text{ 이다.}
 \end{aligned}$$

$$\begin{aligned}
 20. \sum_{n=0}^{\infty} \frac{x^n}{n!} &= e^x \\
 \Rightarrow \sum_{n=0}^{\infty} \frac{x^{n+1}}{n!} &= xe^x \\
 \Rightarrow \int_0^1 \sum_{n=0}^{\infty} \frac{x^{n+1}}{n!} dx &= \int_0^1 xe^x dx = [xe^x - e^x]_0^1 = 1
 \end{aligned}$$

$$\begin{aligned}
 21. \int_0^{\infty} \left(x - \frac{x^3}{2} + \frac{x^5}{2 \cdot 4} - \frac{x^7}{2 \cdot 4 \cdot 6} + \dots\right) dx \\
 = \int_0^{\infty} \left(x - \frac{x^3}{2 \times 1} + \frac{x^5}{2^2 \times (1 \times 2)} - \frac{x^7}{2^3 \times (1 \times 2 \times 3)} + \dots\right) dx \\
 = \int_0^{\infty} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2^n n!} dx = \int_0^{\infty} x \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{x^2}{2}\right)^n dx \\
 = \int_0^{\infty} x e^{-\frac{x^2}{2}} dx = \left[-e^{-\frac{x^2}{2}}\right]_0^{\infty} = 1
 \end{aligned}$$

$$\begin{aligned}
 22. \sum_{n=1}^{\infty} \frac{x^n}{n} &= -\ln(1-x) \\
 \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n 2^{n+1}} &= \sum_{n=1}^{\infty} \frac{1}{n} \cdot \left(\frac{1}{2^2}\right)^n \cdot \frac{1}{2} \\
 &= \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n} \left(\frac{1}{4}\right)^n = \frac{1}{2} \left[-\ln\left(1 - \frac{1}{4}\right)\right] \\
 &= \frac{1}{2} \ln \frac{4}{3}
 \end{aligned}$$

$$\begin{aligned}
 23. \sin x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \\
 \Rightarrow \sum_{n=0}^{\infty} \frac{(-1)^n \pi^{2n+1}}{4^{2n+1} (2n+1)!} &= \sum_{n=0}^{\infty} (-1)^n \frac{\left(\frac{\pi}{4}\right)^{2n+1}}{(2n+1)!} = \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}
 \end{aligned}$$

$$\begin{aligned}
 24. \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n} &= \ln(1+x) \\
 \Rightarrow \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n 2^n} &= \ln \frac{3}{2}
 \end{aligned}$$

$$\begin{aligned}
 25. \tan^{-1} x &= \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} x^{2k+1} \\
 \Rightarrow \frac{\tan^{-1} x}{x} &= \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} x^{2k} \\
 \Rightarrow \frac{\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)}{\frac{1}{\sqrt{3}}} &= \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} \left(\frac{1}{\sqrt{3}}\right)^{2k} \\
 \Rightarrow \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)3^k} &= \frac{\sqrt{3}}{6} \pi
 \end{aligned}$$

$$\begin{aligned}
 26. \text{ㄱ. } \sum_{n=1}^{\infty} \frac{1}{n!} &= \sum_{n=0}^{\infty} \frac{1}{n!} - 1 = e^1 - 1 : \text{무리수} \\
 \text{ㄴ. } \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} &= \tan^{-1} 1 = \frac{\pi}{4} : \text{무리수} \\
 \text{ㄷ. } \sum_{n=1}^{\infty} \frac{1}{n} \left(1 - \frac{1}{e}\right)^n &= -\ln\left(1 - \left(1 - \frac{1}{e}\right)\right) = 1 : \text{유리수} \\
 \text{ㄹ. } \sum_{n=1}^{\infty} \frac{(-1)^n}{2^{2n}} &= \frac{-\frac{1}{4}}{1 - \left(-\frac{1}{4}\right)} = -\frac{1}{5} : \text{유리수}
 \end{aligned}$$

$$\begin{aligned}
 27. \sum_{n=0}^{\infty} x^n &= \frac{1}{1-x} \\
 \Rightarrow \sum_{n=1}^{\infty} n x^{n-1} &= \frac{1}{(1-x)^2} \\
 \Rightarrow \sum_{n=1}^{\infty} n x^n &= \frac{x}{(1-x)^2} \\
 \Rightarrow \sum_{n=1}^{\infty} n^2 x^{n-1} &= \frac{1+x}{(1-x)^3} \\
 \Rightarrow \sum_{n=1}^{\infty} n^2 x^n &= \frac{x+x^2}{(1-x)^3}
 \end{aligned}$$

$$28. \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

$$\Rightarrow \sum_{n=1}^{\infty} nx^{n-1} = (1-x)^{-2}$$

$$\Rightarrow \sum_{n=2}^{\infty} n(n-1)x^{n-2} = 2(1-x)^{-3}$$

$$\Rightarrow \sum_{n=2}^{\infty} n(n-1)x^n = 2x^2(1-x)^{-3}$$

$$\text{즉, } \sum_{n=2}^{\infty} n(n-1)\left(\frac{1}{3}\right)^n = 2 \times \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^{-3} = \frac{3}{4}$$

$$29. e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = \sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)!}$$

$$\Rightarrow xe^x = \sum_{n=1}^{\infty} \frac{x^n}{(n-1)!} \text{에서 양변을 } x \text{에 대해 미분하면}$$

$$\Rightarrow (1+x)e^x = \sum_{n=1}^{\infty} \frac{nx^{n-1}}{(n-1)!}$$

$$\Rightarrow (x+x^2)e^x = \sum_{n=1}^{\infty} \frac{nx^n}{(n-1)!} \text{에서 양변을 } x \text{에 대해 미분하면}$$

$$\Rightarrow (1+2x)e^x + (x+x^2)e^x = \sum_{n=1}^{\infty} \frac{n^2 x^{n-1}}{(n-1)!}$$

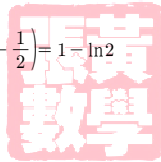
$$\Rightarrow 5e = \sum_{n=1}^{\infty} \frac{n^2}{(n-1)!}$$

$$30. -\ln(1-x) = \sum_{n=1}^{\infty} \frac{x^n}{n} \text{에서 양변을 } x \text{에 대해 적분하면}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{x^{n+1}}{n(n+1)} = (1-x)\ln(1-x) + x$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{x^n}{n(n+1)} = \frac{(1-x)\ln(1-x) + x}{x}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{n(n+1)2^n} = 2\left(\frac{1}{2}\ln\frac{1}{2} - \frac{1}{2}\right) = 1 - \ln 2$$



장황수학

1. 2번	2. 4번	3. 2번	4. 2번	5. 4번
6. 3번	7. 1번	8. 2번	9. 4번	10. 2번
11. 4번	12. 3번	13. 3번	14. 4번	15. 4번

1. $\cos x = -\cos(x - \pi)$

$$= -\left[1 - \frac{1}{2!}(x - \pi)^2 + \frac{1}{4!}(x - \pi)^4 - \dots\right]$$

이므로 $a_0 = -1$, $a_1 = 0$, $a_2 = \frac{1}{2}$ 이다.

$$\text{즉, } a_0 + a_1 + a_2 = -\frac{1}{2}$$

[다른 풀이]

$$\cos x = \sum_{n=0}^{\infty} a_n(x - \pi)^n = a_0 + a_1(x - \pi) + a_2(x - \pi)^2 + \dots$$

$$= f(\pi) + f'(\pi)(x - \pi) + \frac{f''(\pi)}{2!}(x - \pi)^2 + \dots \text{이므로}$$

$f(x) = \cos x$, $f'(x) = -\sin x$, $f''(x) = -\cos x$, ... 에
 $x = \pi$ 를 대입하면,

$$a_0 = f(\pi) = -1, a_1 = f'(\pi) = 0, a_2 = \frac{f''(\pi)}{2!} = \frac{1}{2} \text{이다.}$$

$$\text{즉, } a_0 + a_1 + a_2 = -\frac{1}{2}$$

$$2. f(x) = \frac{5}{2x+3} = \frac{5}{5+2(x-1)} = \frac{1}{1+\frac{2}{5}(x-1)}$$

$$= 1 - \frac{2}{5}(x-1) + \frac{2^2}{5^2}(x-1)^2 - \dots$$

이므로 2차항의 계수는 $\frac{4}{25}$ 이다.

[다른 풀이]

$$f(x) = a_0 + a_1(x-1) + a_2(x-1)^2 + \dots$$

$$= f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \dots \text{이므로}$$

$$f(x) = \frac{5}{2x+3} = 5(2x+3)^{-1}, f'(x) = -10(2x+3)^{-2},$$

$f''(x) = 40(2x+3)^{-3}$, ... 에 $x=1$ 를 대입하면

$$a_2 = \frac{f''(1)}{2!} = \frac{1}{2} \times 40 \times \frac{1}{5^3} = \frac{4}{25} \text{이다.}$$

즉, 2차항의 계수는 $\frac{4}{25}$ 이다.

3. $f(x) = x \sin 2x$ 이라 할 때, $x - \frac{\pi}{6} = t$ 라고 치환하면

$$\begin{aligned} f(t) &= \left(t + \frac{\pi}{6}\right) \sin\left(2t + \frac{\pi}{3}\right) \\ &= \left(t + \frac{\pi}{6}\right) \left(\sin 2t \cos \frac{\pi}{3} + \cos 2t \sin \frac{\pi}{3}\right) \\ &= \left(t + \frac{\pi}{6}\right) \left(\frac{1}{2} \sin 2t + \frac{\sqrt{3}}{2} \cos 2t\right) \\ &= \left(t + \frac{\pi}{6}\right) \left(t - \frac{2}{3}t^3 + \dots + \frac{\sqrt{3}}{2} - \sqrt{3}t^2 + \dots\right) \text{이므로} \\ &= \frac{\sqrt{3}}{12} \pi + \left(\frac{\sqrt{3}}{2} + \frac{\pi}{6}\right)t + \dots + \left(-\sqrt{3} - \frac{\pi}{9}\right)t^3 + \dots \text{이다.} \end{aligned}$$

그러므로

$$a_1 + 6a_3 = \frac{\sqrt{3}}{2} + \frac{\pi}{6} + 6\left(-\sqrt{3} - \frac{\pi}{9}\right) = \frac{-11\sqrt{3}}{2} + \frac{-\pi}{2} \text{이고,}$$

$$A = -\frac{11}{2}, B = -\frac{1}{2} \text{이므로 } A + B = -6 \text{이다.}$$

[다른 풀이]

$f(x) = x \sin 2x$ 일 때,

$$f'(x) = \sin 2x + 2x \cos 2x,$$

$$f''(x) = 2\cos 2x + 2\cos 2x - 4x \sin 2x \\ = 4\cos 2x - 4x \sin 2x$$

$$f^{(3)}(x) = -8\sin 2x - 4\sin 2x - 8x \cos 2x \\ = -12\sin 2x - 8x \cos 2x$$

이고 $x = \frac{\pi}{6}$ 에서 Taylor 급수를 이용하면,

$$a_1 = f'\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2} + 2 \times \frac{\pi}{6} \times \frac{1}{2} = \frac{\sqrt{3}}{2} + \frac{\pi}{6},$$

$$a_3 = \frac{f^{(3)}\left(\frac{\pi}{6}\right)}{3!} \text{이므로 } 6a_3 = -12 \times \frac{\sqrt{3}}{2} - 8 \times \frac{\pi}{6} \times \frac{1}{2} = -6\sqrt{3} - \frac{2}{3}\pi \text{이다.}$$

$$\text{즉, } a_1 + 6a_3 = \frac{-11\sqrt{3}}{2} + \frac{-\pi}{2} \text{이므로 } A + B = -6 \text{이다.}$$

$$4. f(x) = \ln(1 + \sin x) = \sin x - \frac{1}{2}\sin^2 x + \frac{1}{3}\sin^3 x - \frac{1}{4}\sin^4 x + \dots$$

$$= \left(x - \frac{1}{3!}x^3 + \dots\right) - \frac{1}{2}\left(x - \frac{1}{3!}x^3 + \dots\right)^2 + \frac{1}{3}\left(x - \frac{1}{3!}x^3 + \dots\right)^3 - \dots$$

$$= x - \frac{1}{2}x^2 + \frac{1}{6}x^3 + \dots$$

$$\text{이므로 } a_0 + a_1 + a_2 + a_3 = 0 + 1 - \frac{1}{2} + \frac{1}{6} = \frac{2}{3} \text{이다.}$$

$$5. f(x) = \frac{x}{(2+x)(2-x)} = \frac{x}{4-x^2} = \frac{\frac{x}{4}}{1-\frac{x^2}{4}} \\ = \frac{x}{4} \left(1 + \frac{x^2}{4} + \frac{x^4}{16} + \dots\right) = \frac{x}{4} + \frac{x^3}{16} + \dots$$

$$\text{이므로 } f^{(3)}(0) = \frac{3!}{16} = \frac{3}{8} \text{이다.}$$

$$6. f(x) = \frac{x}{1-x+x^2} = \frac{(1+x)x}{(1+x)(1-x+x^2)} = \frac{x+x^2}{1+x^3} \\ = (x+x^2)(1-x^3+x^6-x^9+\dots) \\ = \dots + x^7 + x^8 + \dots$$

$$\text{이므로 } f^{(6)}(0) + f^{(7)}(0) + f^{(8)}(0) = 7! + 8! \text{이다.}$$

$$7. f(x) = \sin(\tan^{-1}(x^2)) = \sin \alpha \quad (\because \tan^{-1} x^2 = \alpha)$$

$$= \frac{x^2}{\sqrt{1+x^4}} = x^2(1+x^4)^{-\frac{1}{2}}$$

$$= x^2 \left(1 - \frac{1}{2}x^4 + \dots\right) = x^2 - \frac{1}{2}x^6 + \dots$$

$$\text{이므로 } f^{(6)}(0) = -\frac{6!}{2} = -\frac{6!}{2} \text{이다.}$$

8. $\sin 3x = 3\sin x - 4\sin^3 x$ 이므로

$$\begin{aligned} \sin^3 x &= \frac{1}{4}(3\sin x - \sin 3x) \\ &= \frac{3}{4}\left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots\right) \\ &\quad - \frac{1}{4}\left((3x) - \frac{1}{3!}(3x)^3 + \frac{1}{5!}(3x)^5 - \frac{1}{7!}(3x)^7 + \dots\right) \\ &= \dots + \frac{1}{4}\left(\frac{3}{5!} - \frac{3^5}{5!}\right)x^5 + \dots \text{이다.} \end{aligned}$$

$$\text{즉, } f^{(5)}(0) = \frac{1}{4}\left(\frac{3}{5!} - \frac{3^5}{5!}\right) \times 5! = \frac{1}{4}(3 - 3^5) = \frac{3}{4}(1 - 81) = -60$$

[다른 풀이]

$$\begin{aligned} f(x) &= \sin^3 x = \sin x \sin^2 x = \sin x \left(\frac{1 - \cos 2x}{2} \right) \\ &= \left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \dots \right) \left[\frac{1}{2} - \frac{1}{2} \left(1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \dots \right) \right] \\ &= \left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \dots \right) \left(x^2 - \frac{1}{3}x^4 + \dots \right) \\ &= \dots + \left(-\frac{1}{3} - \frac{1}{6} \right) x^5 + \dots \end{aligned}$$

이므로 $f^{(5)}(0) = -\frac{1}{2} \times 5! = -60$ 이다.

9. $\sum_{n=0}^{\infty} \frac{(x-a)^{n+1}}{2^{2n+3}}$ 일 때, 비율판정법을 적용하면

$$\begin{aligned} \frac{1}{4} |x-a| < 1 &\Rightarrow |x-a| < 4 \\ \Rightarrow -4 < x-a < 4 &\Rightarrow -4+a < x < 4+a \text{ 이고} \\ \text{수렴구간이 } (-1, 7) &\text{이므로 } a=3 \text{이다.} \end{aligned}$$

$$\begin{aligned} \text{이 때, } f(x) &= \sum_{n=0}^{\infty} \frac{(x-3)^{n+1}}{2^{2n+3}} \text{ 이므로} \\ f(6) &= \sum_{n=0}^{\infty} \frac{3^{n+1}}{2^{2n+3}} = \frac{3}{8} \sum_{n=0}^{\infty} \left(\frac{3}{4} \right)^n = \frac{3}{8} \times \frac{1}{1 - \frac{3}{4}} = \frac{3}{2} \text{이다.} \end{aligned}$$

10. $\ln(1+x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1} x^{n+1}$ 이므로 $\ln(1+x) - x = \sum_{n=1}^{\infty} \frac{(-1)^n}{n+1} x^{n+1}$

$$\text{이고 } \frac{\ln(1+x) - x}{x} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n+1} x^n \text{이다.}$$

$$\text{즉, } \sum_{n=1}^{\infty} \frac{(-1)^n}{n+1} \left(\frac{1}{3} \right)^n = \frac{\ln \frac{4}{3} - \frac{1}{3}}{\frac{1}{3}} = 3 \ln \frac{4}{3} - 1$$

11. $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} = (1-x)^{-1}$ 이므로 미분하면

$$\sum_{n=1}^{\infty} nx^{n-1} = (1-x)^{-2} \text{이고, } \sum_{n=2}^{\infty} n(n-1)x^{n-2} = 2(1-x)^{-3} \text{이다.}$$

$$\text{즉, } \sum_{n=2}^{\infty} n(n-1) \left(\frac{1}{2} \right)^{n-2} = 2 \times \left(\frac{1}{2} \right)^{-3} = 16$$

12. $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ 이므로 $x^2 e^x = \sum_{n=0}^{\infty} \frac{x^{n+2}}{n!}$ 이고

$$\text{미분하면 } 2xe^x + x^2 e^x = \sum_{n=0}^{\infty} \frac{(n+2)x^{n+1}}{n!} \text{이다.}$$

$$\text{또한 미분하면 } 2e^x + 4xe^x + x^2 e^x = \sum_{n=0}^{\infty} \frac{(n+2)(n+1)x^n}{n!} \text{이다.}$$

$$\text{따라서 } x=1 \text{을 대입하면 } \sum_{n=0}^{\infty} \frac{(n+2)(n+1)}{n!} = 2e + 4e + e = 7e \text{이다.}$$

13. $-\ln(1-x) = \sum_{n=1}^{\infty} \frac{1}{n} x^n$ 을 적분하면

$$-\int \ln(1-x) dx = \sum_{n=1}^{\infty} \frac{1}{n(n+1)} x^{n+1} + C_1$$

$$\Rightarrow (1-x) \ln(1-x) + x + C_2 = \sum_{n=1}^{\infty} \frac{1}{n(n+1)} x^{n+1} + C_1$$

$$\Rightarrow (1-x) \ln(1-x) + x = \sum_{n=1}^{\infty} \frac{1}{n(n+1)} x^{n+1} + C \text{이다.}$$

이때, $x=0$ 을 대입하면 $C=0$ 이다.

$$\text{따라서 } (1-x) \ln(1-x) + x = \sum_{n=1}^{\infty} \frac{1}{n(n+1)} x^{n+1} \text{이므로}$$

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} x^n = \frac{(1-x) \ln(1-x) + x}{x} \text{이다.}$$

$$\text{즉, } \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n(n+1)} \left(\frac{1}{3} \right)^n = - \sum_{n=1}^{\infty} \frac{1}{n(n+1)} \left(-\frac{1}{3} \right)^n$$

$$= - \frac{\frac{4}{3} \ln \frac{4}{3} - \frac{1}{3}}{-\frac{1}{3}} = 4 \ln \frac{4}{3} - 1$$

14. $\sum_{n=1}^{\infty} a_n \frac{\pi^n}{6^n n!} = -\frac{\pi}{6 \times 1!} - \frac{\pi^2}{6^2 \times 2!} + \frac{\pi^3}{6^3 \times 3!} + \frac{\pi^4}{6^4 \times 4!} + \dots$

$$= - \left(\frac{\pi}{6 \times 1!} - \frac{\pi^3}{6^3 \times 3!} + \dots \right) + \left(-\frac{\pi^2}{6^2 \times 2!} + \frac{\pi^4}{6^4 \times 4!} - \dots \right)$$

이므로

$$(1) \sin \frac{\pi}{6} = \frac{\pi}{6} - \frac{1}{3!} \left(\frac{\pi}{6} \right)^3 + \frac{1}{5!} \left(\frac{\pi}{6} \right)^5 + \dots$$

$$= \frac{\pi}{6 \times 1!} - \frac{\pi^3}{6^3 \times 3!} + \frac{\pi^5}{6^5 \times 5!} - \dots$$

$$(2) \cos \frac{\pi}{6} - 1 = -\frac{1}{2!} \left(\frac{\pi}{6} \right)^2 + \frac{1}{4!} \left(\frac{\pi}{6} \right)^4 - \frac{1}{6!} \left(\frac{\pi}{6} \right)^6 + \dots$$

$$= -\frac{\pi^2}{6^2 \times 2!} + \frac{\pi^4}{6^4 \times 4!} - \frac{\pi^6}{6^6 \times 6!} + \dots \text{이다.}$$

$$\text{즉, } \sum_{n=1}^{\infty} a_n \frac{\pi^n}{6^n n!} = -\sin \frac{\pi}{6} + \cos \frac{\pi}{6} - 1 = \frac{-3 + \sqrt{3}}{2}$$

15. $x > 0$ 에 대하여

$$\frac{3e^x - 3e^{-x} - 6x - x^3}{x^5} = \frac{6 \sinh x - 6x - x^3}{x^5}$$

$$= \frac{6 \left(x + \frac{1}{3!}x^3 + \frac{1}{5!}x^5 + \dots \right) - 6x - x^3}{x^5}$$

$$= \frac{6}{5!} + \frac{6}{7!}x^2 + \dots > \frac{6}{5!} \text{이므로 } a \text{의 최댓값은 } \frac{1}{20} \text{이다.}$$

1. 3번	2. 1번	3. 4번	4. 1번	5. 1번
6. 4번	7. 2번	8. 2번	9. 2번	10. 2번
11. 5번	12. 2번			

$$\begin{aligned}
 1. \quad f'(0) &= \lim_{h \rightarrow 0} \frac{h^3 \tan^{-1} \frac{1}{h} - 0}{h} = \lim_{h \rightarrow 0} h^2 \tan^{-1} \frac{1}{h} = 0 \\
 \Rightarrow f'(x) &= \begin{cases} 3x^2 \tan^{-1} \frac{1}{x} - \frac{x^3}{1+x^2}, & x \neq 0 \\ 0, & x = 0 \end{cases}, (x \neq 0) \\
 \Rightarrow f''(0) &= \lim_{h \rightarrow 0} \frac{3h^2 \tan^{-1} \frac{1}{h} - \frac{h^3}{1+h^2}}{h} \\
 &= \lim_{h \rightarrow 0} 3h \tan^{-1} \frac{1}{h} - \frac{h^2}{1+h^2} = 0
 \end{aligned}$$

$$2. \quad \sum_{n=1}^{\infty} (\cos 1)^n = \frac{\cos 1}{1 - \cos 1} \text{에서 } \cos 1 \text{의 정확한 값을 계산하기 어렵기}$$

때문에, 1에 가까우면서 계산이 쉬운 각인 $\frac{\pi}{3}$ 을 이용해 값을 추정하면

$$\frac{\pi}{3} \approx 1.047 \text{이므로 } 1 \text{과 매우 가까운 값이고, } \frac{\cos 1}{1 - \cos 1} \approx \frac{\frac{1}{2}}{1 - \frac{1}{2}} = 1 \text{이다.}$$

$$① \quad \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$$

$$② \quad \lim_{n \rightarrow \infty} \arctan n = \frac{\pi}{2} \neq 0 \text{이므로 } \sum_{n=1}^{\infty} \arctan n \text{은 발산한다.}$$

$$③ \quad \sum_{n=1}^{\infty} \frac{2^n}{n!} = e^2 - 1 \approx 6.39$$

$$④ \quad \sum_{n=1}^{\infty} \frac{(-1)^n 2^n}{n!} = \sum_{n=1}^{\infty} \frac{(-2)^n}{n!} = e^{-2} - 1 \approx -0.86$$

$$\begin{aligned}
 3. \quad (\gamma) \quad \int_0^1 x F(x) dx &= \int_0^1 \sum_{k=0}^{\infty} C_k x^{k+1} dx \\
 &= \sum_{k=0}^{\infty} \int_0^1 C_k x^{k+1} dx \\
 &= \sum_{k=0}^{\infty} \left[\frac{C_k}{k+2} x^{k+2} \right]_0^1 = \sum_{k=0}^{\infty} \frac{C_k}{k+2} \quad (\text{참})
 \end{aligned}$$

$$(\iota) \quad F'(x) = \sum_{k=0}^{\infty} \frac{d}{dx} (C_k x^k) = \sum_{k=0}^{\infty} k C_k x^{k-1}$$

$$\Rightarrow x F'(x) = \sum_{k=0}^{\infty} k C_k x^k$$

$$\therefore \sum_{k=0}^{\infty} k 2^k C_k = 2 F'(2) \quad (\text{참})$$

$$(\epsilon) \quad F'(x) = \sum_{k=0}^{\infty} k C_k x^{k-1}$$

$$\Rightarrow F''(x) = \sum_{k=0}^{\infty} k(k-1) C_k x^{k-2}$$

$$\Rightarrow x^2 F''(x) = \sum_{k=0}^{\infty} k(k-1) C_k x^k$$

$$\therefore \sum_{k=0}^{\infty} k(k-1) C_k 2^k = 4 F''(2) \quad (\text{참})$$

$$\begin{aligned}
 4. \quad \sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} \\
 \Rightarrow x \sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+2}}{(2n+1)!} \\
 \Rightarrow \sin x + x \cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n (2n+2) x^{2n+1}}{(2n+1)!} \\
 &= 2x \sum_{n=0}^{\infty} \frac{(-1)^n (n+1) x^{2n}}{(2n+1)!} \\
 \Rightarrow x = \pi : \sin \pi + \pi \cos \pi &= 2\pi \sum_{n=0}^{\infty} \frac{(-1)^n (n+1) \pi^{2n}}{(2n+1)!} \\
 \therefore \sum_{n=0}^{\infty} \frac{(-1)^n (n+1) \pi^{2n}}{(2n+1)!} &= -\frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 5. \quad \sum_{n=1}^{\infty} n^2 x^n &= \frac{x+x^2}{(1-x)^3} \\
 \Rightarrow \sum_{n=1}^{\infty} n^3 x^{n-1} &= \frac{x^2+4x+1}{(1-x)^4} \\
 \Rightarrow \sum_{n=1}^{\infty} n^3 x^n &= \frac{x^3+4x^2+x}{(1-x)^4} \\
 \Rightarrow \sum_{n=1}^{\infty} \frac{n^3}{3^n} &= \frac{33}{8}
 \end{aligned}$$

$$\begin{aligned}
 6. \quad \int_0^{\sqrt{\frac{\pi}{2}}} x^{4n+3} dx &= \left[\frac{1}{4n+4} x^{4n+4} \right]_0^{\sqrt{\frac{\pi}{2}}} = \frac{1}{4n+4} \left(\frac{\pi^2}{4} \right)^{n+1} \\
 \Rightarrow \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \int_0^{\sqrt{\frac{\pi}{2}}} x^{4n+3} dx &= \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \frac{1}{(2n+2)} \left(\frac{\pi^2}{4} \right)^{n+1} \\
 &= \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+2)!} \left(\frac{\pi^2}{2} \right)^{n+1} = \frac{1}{2} \left[1 - \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \left(\frac{\pi^2}{2} \right)^n \right] \\
 &= \frac{1}{2} (1 - \cos \frac{\pi}{2}) = \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 7. \quad \sec x &= 1 + \frac{1}{2} x^2 + \frac{5}{24} x^4 + \dots \text{이므로 } f^{(4)}(x) = 5 + \dots \text{이다.} \\
 \therefore f^{(4)}(0) &= 5
 \end{aligned}$$

$$\begin{aligned}
 8. \quad S(x) &= \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases} = \begin{cases} 1 - \frac{1}{3!} x^2 + \dots, & x \neq 0 \\ 0, & x = 0 \end{cases} \\
 &= 1 - \frac{1}{3!} x^2 + \frac{1}{5!} x^4 - \frac{1}{7!} x^6 + \frac{1}{9!} x^8 - \frac{1}{11!} x^{10} + \dots \\
 \Rightarrow S^{(8)}(0) + S^{(9)}(0) + S^{(10)}(0) &= \frac{8!}{9!} - \frac{10!}{11!} = \frac{1}{9} - \frac{1}{11} = \frac{2}{99}
 \end{aligned}$$

$$\begin{aligned}
 9. \quad \sum_{n=2}^{\infty} (1+c)^{-n} &= \sum_{n=0}^{\infty} (1+c)^{-n} - 1 - (1+c)^{-1} \\
 &= \frac{1}{1 - \frac{1}{1+c}} - 1 - \frac{1}{1+c} \\
 &= \frac{1}{c(1+c)} = 2
 \end{aligned}$$

$$\Leftrightarrow 2c^2 + 2c - 1 = 0 \text{이므로 } c = \frac{-1 \pm \sqrt{3}}{2}$$

$$\text{즉, } \left| \frac{1}{1+c} \right| < 1 \text{을 만족하는 } c \text{는 } \frac{-1 + \sqrt{3}}{2} \text{이다.}$$

10. $a_{2n} = \int_{(2n-1)\pi}^{2n\pi} |e^{-x} \sin x| dx$ 이므로 $f(x) = e^{-x}$ 는 항상 $f(x) > 0$ 이고,

$g(x) = \sin x$ 는 구간 $((2n-1)\pi, 2n\pi)$ 에서 항상 $g(x) < 0$ 이다.

$$\Rightarrow a_{2n} = - \int_{(2n-1)\pi}^{2n\pi} e^{-x} \sin x dx = - \left[\frac{e^{-x}}{2} (-\sin x - \cos x) \right]_{(2n-1)\pi}^{2n\pi} \\ - \frac{1}{2} [-e^{-2n\pi} - e^{-(2n-1)\pi}] = \frac{1}{2} \left(\frac{1}{e^{(2n-1)\pi}} + \frac{1}{e^{2n\pi}} \right)$$

$$\Rightarrow \sum_{n=1}^{\infty} a_{2n} = \sum_{n=1}^{\infty} \frac{1}{2} \left(\frac{1}{e^{(2n-1)\pi}} + \frac{1}{e^{2n\pi}} \right) \\ = \frac{1}{2} \left(\frac{1}{e^{\pi}} + \frac{1}{e^{2\pi}} + \frac{1}{e^{3\pi}} + \dots \right) \\ = \frac{1}{2e^{\pi}} \left(1 + \frac{1}{e^{\pi}} + \frac{1}{e^{2\pi}} + \dots \right) \\ = \frac{1}{2e^{\pi}} \left(\frac{1}{1 - \frac{1}{e^{\pi}}} \right) = \frac{e^{\pi} + 1}{2(e^{2\pi} - 1)}$$

11. $f(x) = \frac{2e^x - 2 - 2x - x^2}{x^3}$

$$= \frac{2 \left(1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \dots \right) - 2 - 2x - x^2}{x^3} \\ = \frac{1}{3} + \frac{1}{4!}x + \dots$$

$x > 0$ 에서 $f(x) = \frac{1}{3} + \frac{1}{4!}x + \dots > \frac{1}{3}$ 이므로

$f(x) > a$ 를 만족하는 a 의 최댓값은 $\frac{1}{3}$ 이다.

12. $G(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$

$$\Rightarrow 2xG(x) = \sum_{n=0}^{\infty} 2a_n x^{n+1}$$

$$= 2a_0 x + 2a_1 x^2 + 2a_2 x^3 + 2a_3 x^4 + \dots$$

$$\Rightarrow 3x^2 G(x) = \sum_{n=0}^{\infty} 3a_n x^{n+2} = 3a_0 x^2 + 3a_1 x^3 + 3a_2 x^4 + \dots$$

$$\Rightarrow G(x) - 2xG(x) - 3x^2 G(x) = a_0 + (a_1 - 2a_0)x + (a_2 - 2a_1 - 3a_0)x^2 \\ + (a_3 - 2a_2 - 3a_1)x^3 + \dots$$

$a_0 = 1, a_1 = 1, a_{n+2} = 2a_{n+1} + 3a_n$ 이므로

$$\Rightarrow G(x) - 2xG(x) - 3x^2 G(x) = 1 - x$$

$$\Rightarrow G(x) = \sum_{n=0}^{\infty} a_n x^n = \frac{1-x}{1-2x-3x^2}$$

장황수학